SITHUM MANUSHA

QUALITY ASSURANCE (QA) INTERN

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ABOUT ME

A third-year Information Technology & Management undergraduate at the University of Moratuwa specializing in Software Quality Assurance, Test Automation, and System Reliability. Experienced in architecting automated E2E testing suites, API verification, and continuous testing pipelines using Playwright (TypeScript), Pytest, JUnit 5, Vitest, and Postman. Demonstrated track record of authoring 86+ automated tests, validating distributed REST microservices, verifying database integrity with SQL, and integrating test suites into GitHub Actions CI/CD workflows. Seeking a Quality Assurance Internship to drive high software quality, defect prevention, and reliable test automation across enterprise environments.

EDUCATION

University of Moratuwa (UOM) - Moratuwa, Sri Lanka **2024 - Present**
B.Sc. (Hons) in Information Technology & Management **(3rd Year | CGPA: 3.23 / 4.0)**

Open University of Sri Lanka (OUSL) - Nawala, Sri Lanka **2025 - Present**
Bachelor of Laws Honours Degree Programme (LL.B.)

St. Joseph's Vaz College - Wennappuwa, Sri Lanka **2009 - 2022**
G.C.E. Advanced Level (2 A's & 1 B) | G.C.E. Ordinary Level (8 A's & 1 B)

FEATURED QA & TESTING PROJECTS

1. AutomationExercise QA Framework — Playwright Test Automation Suite (Individual Project) **2026**

- Developed a comprehensive, production-grade automated end-to-end (E2E) testing suite in **TypeScript** using **Playwright**, structured under the **Page Object Model (POM)** design pattern for maintainability and scalability.
- Implemented custom network route interception to detect and block intrusive third-party ad networks, achieving 100% deterministic, zero-flakiness test runs.
- Engineered dynamic test data pipelines with unique timestamp identifiers and automated test runs via **GitHub Actions CI/CD** with rich HTML reports, failure traces, and video recordings.
- **Technologies:** Playwright, TypeScript, Page Object Model (POM), GitHub Actions (CI/CD), Automated E2E Testing, HTML Reporting, Trace Viewer
- **GitHub:** github.com/SithumManusha/AutomationExercise-Playwright-TS

2. CareerFlow — ATS Platform with 26-Test Automated Multi-Tier Testing Suite (Individual Project) **2026**

- Implemented a multi-tier automated test harness comprising **26 automated tests** covering **JUnit 5** (Spring Boot gateway), **Pytest** (FastAPI NLP worker), and **Playwright** (Next.js UI).
- Validated NLP keyword extraction and Cosine Similarity scoring algorithms using boundary value analysis, equivalence partitioning, and negative test scenarios.
- Conducted API stress testing and rate-limit verification (10 req/min thresholds) with automated regression test execution inside **Docker Compose** and **GitHub Actions CI/CD**.
- **Technologies:** JUnit 5, Pytest, Playwright, Postman, Java 21, Spring Boot 3, Python 3.12, FastAPI, Next.js 14, Docker, GitHub Actions
- **Live Demo:** careerflow-pink.vercel.app | **GitHub:** github.com/SithumManusha/careerflow

3. Ledgerly — Financial Platform with 60+ Vitest Automated Unit Tests (Individual Project) **2026**

- Built and maintained a comprehensive automated unit test suite with **60+ automated tests (Vitest)** ensuring 100% calculation accuracy for category budget health scores and 6-month balance savings projections.
- Formulated test cases for group expense splitting across 4 distribution methods, validating mathematical edge cases, precision roundoffs, and automated PDF export statements.
- Performed API endpoint verification, schema validation using **Zod**, and regression testing across a type-safe stack (**React, TypeScript, Node.js, PostgreSQL**).
- **Technologies:** Vitest, Automated Unit Testing, Zod Schema Validation, Postman, React, TypeScript, Node.js, PostgreSQL, Render
- **Live Demo:** ledgerly-mbcd.onrender.com | **GitHub:** github.com/SithumManusha/Ledgerly

4. Super POS — Multi-Branch Retail System Sync & Resilience Testing (Group Project) **2026**

- Performed integration, offline resilience, and data synchronization testing for client-side offline sales buffering (**useOfflineSync** with **IndexedDB**) and auto-reconnection to **PostgreSQL**.
- Tested **Last-Write-Wins (LWW)** conflict detection and validated interactive 3-way resolver behavior (Trust Local, Trust Cloud, Smart Merge) under simulated high-concurrency race conditions.
- Executed WebSocket reliability and heartbeat testing via **Socket.io**, verifying fleet broadcast topology, P50/P95/P99 latency benchmarks, and audit trail PDF exports.
- **Technologies:** Integration Testing, Postman API Testing, Socket.io, IndexedDB, NestJS, Node.js, Prisma ORM, PostgreSQL (Supabase), Turborepo
- **Live Demo:** pos-system-pos-web-app.vercel.app | **GitHub:** github.com/Ryzera/Ryzera-POS

TECHNICAL SKILLS

- **Test Automation & Frameworks:** Playwright (TypeScript), Vitest, Pytest, JUnit 5, Page Object Model (POM), E2E Testing, Unit Testing, Integration Testing, Regression Testing
- **API & Performance Testing:** Postman API Testing, Swagger / OpenAPI, RESTful API Validation, WebSocket Testing (Socket.io), Rate Limit Verification
- **QA Methodologies & Quality:** SDLC, STLC, Defect Tracking, Test Case Design, Boundary Value Analysis, Equivalence Partitioning, Bug Life Cycle, Jira, Agile / Scrum
- **Programming Languages:** TypeScript, JavaScript, Python, Java, SQL, C
- **CI/CD & DevOps:** GitHub Actions (CI/CD Automated Test Runners), Docker, Docker Compose, Git, Turborepo, Render Cloud
- **Databases & Data Verification:** PostgreSQL (Supabase), MySQL, Redis, Database Query Verification (SQL)

CORE QA COMPETENCIES & ARCHITECTURE

Test Automation Architecture, E2E Browser Testing, API Verification, Test Case Authoring, Defect Lifecycle Management, Agile/Scrum Quality Practices, CI/CD Automated Test Execution, Data Integrity Verification

CERTIFICATIONS

- **Software Testing with Generative AI** — Simplilearn
- **AWS Cloud Practitioner Essentials** — Amazon Web Services (AWS)
- **Introduction to Artificial Intelligence** — Simplilearn
- **Introduction to Cybersecurity** — Cisco Networking Academy

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Member of the Organizing Committee, Official Career Fair conducted by University of Moratuwa (2026)
- Member of the Organizing Committee, Project "Rest with Hope" conducted by Leo Club, UoM (2025)
- Active Member — FIT Moments (Faculty Media Body) | Leo Club of UoM | Legal Youth Community Sri Lanka

LANGUAGES

- English: Professional Working Proficiency (Oral & Written) | Sinhala: Native Proficiency

REFERENCES

Dr. A.L.A.R.R. Thanuja (B.Sc. (Sri Lanka), Ph.D. (USA)),
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